

Bléum Nails By Micah

Villanueva Bryan O. - Leader

Regondola, Jezreel P.

Ruperto, April Anne A.

Sorellano, John Kenneth T.

Talagtag, Mark Angel T.

.Villacin, Justine R.

University of Caloocan City

Caloocan City

December 6, 2025

Approval Sheet

This design project entitled “**Bléum Nails By Micah**” prepared by **Regondola, Jezreel P., Ruperto, April Anne A., Sorellano, John Kenneth T., Talagtag, Mark Angel T., Villacin, Justine R., Villanueva, Bryan O.** of the Computer Engineering Department, was examined and evaluated by the members of the Student Design Evaluation Panel and is hereby recommended for approval.

ENGR. MARIA RIZZETTE H. SAYO

Adviser

Student Design Evaluation Panel:

ENGR/DR. FIRSTNAME LASTNAME

Panel Member

ENGR/DR. FIRSTNAME LASTNAME

Panel Member

ENGR/DR. FIRSTNAME LASTNAME

Lead Panel

ENGR. MARIA CONCEPCION MIRABUENO

Program Chair

UNIVERSITY OF CALOOCAN CITY
Caloocan City

SOFTWARE DESIGN PROJECT INFORMATION

Semester, SY 2025-2026

Student/Team Team: SENTINAILS	Regondola, Jezreel P. Ruperto, April Anne A. Sorellano, John Kenneth T. Talagtag, Mark Angel T. Villacin, Justine R. Villanueva, Bryan O.
Project Title	BLÉUM NAILS
Project Concentration Area	SOFTWARE DESIGN
Design Objectives	The general objective of the project is to make a software for our client Specifically, it aims to: 1. To analyze the client's requirements and identify the functional and non-functional specifications of the software. 2. To design a user-friendly interface that ensures efficient navigation and usability. 3. To develop the core features of the software in alignment with the client's workflow. 4. To integrate necessary tools or modules such as database systems, reporting features, or automation functions. 5. To test and debug the software to ensure performance, reliability, and security. 6. To deploy the software in the client's environment and ensure smooth implementation. 7. To provide documentation and training so the client can effectively use and maintain the system.
Constraints	
Constraint (Metric)	The software must be compatible with the client's existing hardware device and operating system.
Constraint (Metric)	The project must be completed within the agreed project timeline or deadline.
Constraint (Metric)	The software must comply with data privacy laws
Constraint (Metric)	The software must meet minimum performance requirements

Constraint (Metric)	The system must use specific programming languages, frameworks, or databases required by the client.
Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.	
Constraint	A constraint is a limit or restriction. Hopefully your school won't cut the sports or music programs due to budget constraints! You might notice that constraint is similar to the verb constrict, as in boa constrictor. A constraint tightens around you like a snake and prevents you from doing what you want.
Standards	
Standard 1	Compatibility Standard Constraint: <i>The software must be compatible with the client's existing hardware and operating system.</i> Standard: • The system must undergo hardware device and OS compatibility testing using the client's actual environment. • Only technologies and libraries that support the client's OS version and hardware device specs may be used. • The software must operate without requiring hardware device upgrades unless approved by the client.
Standard 2	Time / Schedule Standard Constraint: <i>The project must be completed within the agreed project timeline or deadline.</i> Standard: • The development team must follow a project timeline with defined milestones (design, development, testing, deployment). • Weekly progress reports must be submitted to track schedule compliance. • Any changes affecting the timeline require approval through a formal change request.
Standard 3	Data Privacy Compliance Standard Constraint: <i>The software must comply with data privacy laws.</i> Standard: • The system must follow relevant data privacy regulations (e.g., Data Privacy Act, GDPR, or local equivalent). • Sensitive data must be encrypted in transit and at rest. • Access to personal or confidential information must require role-based authentication.

	The system must provide audit logs for any access or changes to user data.
Standard 4	Performance Standard Constraint: <i>The software must meet minimum performance requirements.</i> Standard: - Response time for major functions must not exceed the defined threshold (e.g., ≤ 2 seconds). - Load testing must be conducted to ensure performance under expected user volume. - Resource usage (CPU, memory, storage) must stay within acceptable limits based on client hardware.
Standard 5	Technology / Development Standard Constraint: <i>The system must use specific programming languages, frameworks, or databases required by the client.</i> Standard: - Only approved technologies (e.g., Python, Java, React, MySQL) may be used for system development. - All developers must follow the coding conventions for the selected language (e.g., PEP 8, Java Style Guide, Airbnb JS Standards). - Libraries and dependencies must be documented and approved to prevent unauthorized or incompatible tools.

Abstract

This project presents the development of a **Bléum Nails By Micah** Appointment System designed to streamline the scheduling process between clients and the Admin. The software aims to eliminate manual booking errors, reduce waiting times, and enhance overall customer service by providing an organized and user-friendly digital platform. The system enables clients to book appointments, view available time slots, and receive confirmations, while allowing salon staff to efficiently manage schedules and track client information.

The development of the system followed a structured process that included requirements gathering, system design, coding, testing, and deployment. The project adhered to key constraints, including compatibility with the client's existing hardware and operating system, compliance with data privacy laws, fulfillment of performance standards, timely project completion, and the use of specified programming languages and frameworks. Rigorous testing ensured the reliability, usability, and accuracy of the system.

Overall, the Nail Salon Appointment System provides a modern and efficient solution that enhances the operational workflow of the salon and improves the booking experience for clients.

List of Tables

List of figures

List of abbreviation

Definition of terms

TABLE OF CONTENTS

SOFTWARE DESIGN TITLE	1
Approval Sheet	2
Abstract	5
List of Tables	6
List of figures	7
List of abbreviation	8
Definition of terms	9
TABLE OF CONTENTS	10
CHAPTER 1: THE PROJECT AND ITS BACKGROUND	13
1.1 The Problem	13
1.2 The Client	13
1.3 The Project	13
1.4 Project Objectives	13
1.5 Scope and Delimitations	13
1.6 Design Constraints	14
1.7 Engineering Standards	15
1.8 Engineering Design Process	15
CHAPTER 2: SOFTWARE DESIGN	17
2.1 Description of the Design Solution	17
2.1.1 General Description	17
2.1.2 Engineering Principles Involved	17
2.1.3 Prior Art Analysis	17
2.2 General System Architecture	18
2.2.1 Hardware Elements	18
2.2.2 Software Elements	18
2.2.2 System Algorithm	18
2.2.3 Data, Datasets, and Processing	18
2.3 Design Alternatives	19
2.3.1 Rationale for Design Alternatives	19
2.3.1 Design Alternative 1:	19
2.3.2 Design Alternative B	20
2.3.3 Design Alternative C	20
2.4 Standards Involved in the Design	20
	11

CHAPTER 3: DESIGN TRADEOFFS	22
3.1 Summary of Constraints	22
3.2 Trade-offs	22
3.2.1 Tradeoff 1: Constraint A (Metric)	23
3.2.2 Tradeoff 2: Constraint B (Metric)	23
3.2.3 Tradeoff 3: Constraint C (Metric)	24
3.2.4 Tradeoff 4: Constraint D (Metric)	24
3.2.5 Tradeoff 5: Constraint E (Metric)	24
3.3 Summary of the Normalized Values of the Three Designs	24
3.4 Designers Raw Ranking for the Three Designs	24
3.5 Sensitivity Analysis	25
3.6 Influence of the Design Tradeoffs in the Final Design	25
CHAPTER 4: FINAL DESIGN	26
4.1 Final Design	26
4.1.1 Software Application	26
4.1.2 Hardware Design	26
4.2 Test Procedures and Evaluation	26
4.2.1 Test Procedures	26
4.2.2 Test Evaluation	26
4.3 Test and Evaluation Results	26
4.3.1 Test Results	26
4.3.2 Evaluation Results	26
4.4 Conclusion	26
4.5 Impact of the Design	26
4.5.1 Societal	26
4.5.2 Ethical	26
4.5.3 Legal	26
4.6 Sustainability Plan	26
CHAPTER 5: BUSINESS PLAN AND MODEL	27
5.1 Business Plan	27
5.1.1 Executive Summary	27
5.1.2 General Company Description	27
5.1.3 Products and Services Offered	27
5.1.4 Marketing Plan	27

5.1.5 Marketing Strategy	27
5.2 Business Model	27
5.3 Intellectual Property (IP) Reports	27
REFERENCES	28
APPENDICES	29

CHAPTER 1: THE PROJECT AND ITS BACKGROUND

Background of the Project

In the beauty and wellness industry, efficient time management is the cornerstone of business success. **Bléum Nails By Micah**, like many growing service providers, initially operated using manual booking methods relying on physical logs or informal messaging to schedule appointments. While these methods may work for a small volume of clients, they often lead to operational error, such as scheduling overlaps, lack of real time availability for customers, and the potential for lost client data.

As digital literacy among consumers increases, there is a rising expectation for convenient, 24/7 accessible booking platforms. To address these challenges, this project introduces the **Bléum Nails By Micah Appointment System**. This software design is intended to bridge the gap between manual labor and automated efficiency by providing a structured, secure, and user friendly digital environment. By modernizing the scheduling process, the business can reduce administrative overhead and focus more on delivering high quality nail services to its clients.

1.1 The Problem

The primary issue facing **Bléum Nails By Micah** is the reliance on manual booking processes, which are inefficient and prone to human error. These problems directly affect the salon's ability to provide a smooth experience for its customers.

The specific problems identified include:

- **Manual Booking Errors:** Without a digital record, it is easy to double-book a time slot or lose track of a client's appointment.
- **Increased Waiting Times:** Poorly organized schedules result in clients waiting longer than expected, leading to a decrease in customer satisfaction.
- **Operational Inefficiency:** Staff spend too much time manually confirming available slots and managing schedules instead of focusing on nail services.
- **Lack of Real Time Access:** Clients cannot view available time slots instantly, making the booking process slow and inconvenient.

1.2 The Client

The client for this project is **Bléum Nails By Micah**, a professional nail technology business. The business requires a reliable digital platform to interface with its customers and manage its daily service calendar efficiently.

Table 1-1. Client and Engineering Requirements / Considerations

Client Requirements / Considerations	Engineering Requirements / Considerations
The system can allow clients to book appointments and view available time slots.	Implementation of a real time scheduling algorithm and database management for availability tracking.

Client Requirements / Considerations	Engineering Requirements / Considerations
The system can provide the owner with an organized way to manage schedules.	Development of an administrative dashboard with CRUD (Create, Read, Update, Delete) functionalities.
The system can protect client information and maintain data privacy.	Integration of role-based authentication and data encryption standards
The system should generate summaries of business activity	Integration of reporting features or automation functions for administrative use.
The system user interface must be professional and represent the brand.	Application of modern UI/UX principles during the design phase to ensure high usability.

1.3 The Project

The project, titled **Bléum Nails By Micah Appointment System**, is a software design solution developed by **Team Sentinails**. The system acts as a digital bridge between the salon's clients and the administrator. It is designed to replace manual record-keeping with an organized digital platform that simplifies the booking process, manages salon capacity, and tracks client data securely. The development follows a structured engineering design process including requirements gathering, design, coding, testing, and deployment.

1.4 Project Objectives

The general objective of this project is to create the Bléum Nails By Micah Appointment System, a digital platform designed to streamline the salon's scheduling process, eliminate manual booking errors, and improve the overall efficiency of client management.

Specifically, the project aims to:

1. **Design** the software elements required for the system's operation, such as the interface setup for existing client devices.
 - Ensure the design is compatible with the client's current operating systems and hardware.
2. **Develop** the software components of the appointment system, including the booking module and the administrative dashboard.
 - Create a secure database for managing client information and appointment history.
 - Implement a real-time scheduling interface for both clients and salon staff.
3. **Test and evaluate** the system's accuracy to ensure reliability, security, and high performance in a real-world environment.

1.5 Scope and Delimitations

The scope of this project is to develop a user-friendly website for a home-based nail technician to manage bookings and payments more efficiently. Since the system is web-based, it is accessible 24/7, allowing customers to browse services and schedule appointments at any time that is convenient for them. The website will feature a list of nail services, pricing, and a calendar of available time slots. To secure a booking, the system requires a down payment; customers must scan the technician's GCash QR code and upload a photo of their proof of payment (receipt) directly to the website before the appointment can be finalized. To help customers feel confident about the legitimacy of the business, the website will include direct links to the technician's Facebook page, Instagram, and email address.

Regarding the limitations of the website, the system is designed strictly for a single-person, home-based service and does not support large salons with multiple employees or branches. The payment feature is limited to displaying a GCash QR code and does not have an automatic verification system; the technician must manually check the uploaded receipt later. Additionally, the website does not have a built-in chat or messaging feature for the technician and client to talk directly. Instead, any communication or inquiries must be done through the provided social media links or email. While the website is mobile-friendly, it is not a native app available for download in app stores. Finally, the project will not include advanced tools like "virtual nail try-ons" or inventory management for nail supplies, and it will focus on standard security for a small-scale website.

1.6 Design Constraints

1. Primary Constraints: These are used to select the best design.

- **Usability (User Satisfaction Score)**

Usability is how easy it is for the owner and customers to use the system. The User Satisfaction Score measures how satisfied users are with the system on a scale of 1 to 10. It is computed by asking users to rate their experience after using the system and calculating the average of all responses. A higher satisfaction score means the system is easier and more pleasant to use. Therefore, the design with the highest user satisfaction score is the winning design.

- **Functionality (Task Success Rate)**

Functionality is the system's ability to let users complete important tasks like booking appointments and managing schedules. Task Success Rate measures the percentage of users who can successfully complete key tasks without errors. It is computed by observing users attempting tasks and dividing successful completions by total attempts, then multiplying by 100. A higher task success rate

means the system works better and has all needed features. Therefore, the design with the highest task success rate is the winning design.

- **Performance (Page Load Speed)**

Performance is how fast the system responds when users click buttons or load pages. Page Load Speed measures the average time in seconds it takes for a page to fully load. It is computed by recording how long each page takes to load during testing and calculating the average time. Lower load times mean better performance, with pages ideally loading in under 2 seconds on mobile phones. Therefore, the design with the fastest page load speed is the winning design.

- **Reliability (System Availability)**

Reliability is the system's ability to work without crashing or going offline. System Availability measures what percentage of time the system is working properly over a week. It is computed by dividing the hours the system worked correctly by the total hours in a week, then multiplying by 100. Higher availability means users can always access the system when they need it. Therefore, the design with the highest system availability is the winning design.

- **Maintainability (Code Simplicity Score)**

Maintainability is how easy it is for programmers to fix bugs and add new features to the system. Code Simplicity Score measures how complex the programming code is on a scale where lower numbers mean simpler code. It is computed by analyzing the code structure and counting decision points, branches, and complexity patterns. Simpler code with lower scores is easier to update and maintain over time. Therefore, the design with the lowest code simplicity score is the winning design.

2. Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.

- **Accessibility**

All designs work on phones, tablets, and computers with readable text and easy navigation. This basic accessibility is required in every design option.

- **Legal Requirements**

All designs protect customer privacy (names, phone numbers, home addresses) and follow data protection laws. Every option meets these legal requirements equally.

- **Sustainability**

All designs use standard web hosting with similar energy usage. The environmental impact is the same for all options.

- **Cultural and Social Support**

All designs let the nail technician customize their services, hours, and policies to fit their style and community. This flexibility is built into every option.

1.7 Engineering Standards

Expanded and Customized Version: 1.7 Engineering Standards

The engineering standards serve as the foundational framework for the design, development, and operation of the online booking and management system for the nail salon. Adherence to these standards ensures system security, reliability, interoperability, maintainability, and a high-quality user experience.

1.7.1 Web Development & Architecture

- **ISO/IEC 25010:2011 (Systems and software Quality Requirements and Evaluation - SQuaRE):** The system architecture and design shall prioritize usability, reliability, performance efficiency, security, and maintainability as defined by this quality model.
- **W3C Web Standards (HTML5, CSS3, ECMAScript 2023+):** All front-end development shall comply with the latest stable World Wide Web Consortium (W3C) recommendations to ensure cross-browser compatibility, accessibility, and future-proofing.
- **RESTful API Principles:** Back-end services shall be designed following Representational State Transfer (REST) architectural constraints to ensure a stateless, scalable, and modular interface for client-server communication.
- **Progressive Web App (PWA) Criteria:** The application shall implement core PWA technologies (Service Workers, Web App Manifest) to enable reliable performance, offline capabilities, and a native-app-like experience on mobile devices.

1.7.2 Security & Data Protection

- **OWASP Application Security Verification Standard (ASVS) Level 1:** The application shall comply with ASVS Level 1 to protect against common security vulnerabilities, with a specific focus on:
 - Secure authentication and session management (for Admin and Client login).
 - Protection against injection attacks (SQL, XSS).
 - Secure handling of sensitive data (client info, payment details).
- **General Data Protection Regulation (GDPR) / Applicable Local Privacy Laws:** The system shall be designed to facilitate compliance with data privacy regulations. This includes implementing secure data storage, providing mechanisms for data subject requests (e.g., data deletion), and ensuring lawful processing of client information (e.g., contact details, allergy information).
- **Payment Card Industry Data Security Standard (PCI DSS):** Any component handling, processing, or transmitting payment card information shall be designed to facilitate PCI DSS compliance. This mandates the use of certified, third-party payment gateways (e.g., for GCash integration) and ensures no sensitive payment data is stored on the application's primary servers.

1.7.3 User Experience (UX) & Accessibility

- **Web Content Accessibility Guidelines (WCAG) 2.1 Level AA:** The user interface shall conform to WCAG 2.1 Level AA success criteria to ensure the website is accessible to people with a wide range of disabilities, supporting the business goal of being client-friendly.
- **Mobile-First Responsive Design:** The design and development process shall follow a mobile-first responsive approach, ensuring optimal functionality and aesthetics across all device screen sizes (PC and mobile), as per the business requirements.

1.7.4 Operational & Integration Standards

- **iCalendar (RFC 5545):** For calendar functionality and appointment management, the system shall use the iCalendar data format for representing calendar events to ensure interoperability with external calendar applications for reminder generation.
- **PDF/A-1 (ISO 19005-1):** Automatically generated receipts shall be created in the PDF/A-1 format to ensure the long-term preservation and readability of these financial documents.
- **Email Standards (SMTP, RFC 5322):** Automated communications (reminders, confirmations) shall be sent using standard SMTP protocols with properly formatted headers to ensure high deliverability and prevent marking as spam.

1.7.5 Project Management & Development

- **Git Version Control:** All source code shall be managed using the Git version control system, following a feature-branch workflow (e.g., Git Flow) to track changes, enable collaboration, and maintain code integrity.
- **Agile Methodology:** The project shall follow an Agile development framework (e.g., Scrum or Kanban) with iterative sprints, regular client demos, and adaptive planning to accommodate the evolving requirements outlined in the business needs.

1.8 Engineering Design Process

The Engineering Design Process is used to organize a set of procedures and to develop workable solutions to issues. Our team starts with defining the issue and determining the particular requirements of the clients and the possible limitations of the project. After that, our team does background research and brainstorming sessions to produce a range of innovative concepts that will help us to achieve the clients requirements . After that we will create a prototype, which is an early model used for testing, after selecting a potential idea. In order to make sure the finished product is a user friendly and effective as possible, the software will be test, evaluate, and improve based on performance data during the testing phase, which is the most crucial portion of the cycle.

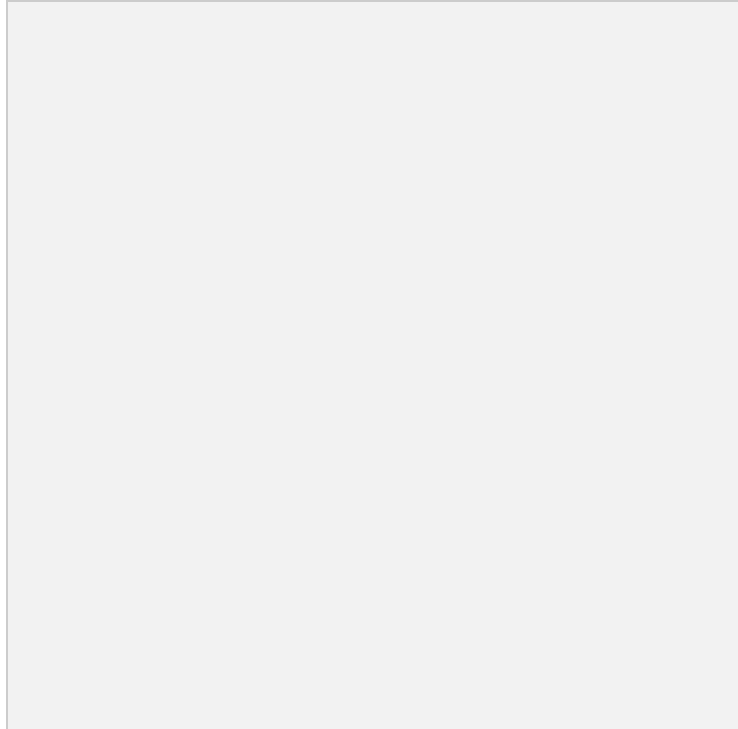


Figure 1.2 The Engineering Design Process (OCTE)

1.8.1 Ask: Identifying the Need and Constraints

In the Engineering Design Process, the project must start with identifying the need and constraints in order to precisely describe the problem and the parameters of the solution. The team determines the client and their particular needs during this "Ask" step to make sure the solution truly resolves a practical problem. Constraints, or non-negotiable limitations like budget, available resources or hardware specification, and size requirements, are identified as early as possible so that the team will avoid spending time on unrealistic concepts and establish a clear path for the remainder of the design process by defining precisely what the product must do and what constraints it must function under.

1.8.2 Research the Problem

In this stage of the Engineering Design process, Before the team attempts to create something new, They seek outward to comprehend the current environment or they find other existing projects during the Research the Problem stage. This stage entails looking into how other people have resolved comparable issues in the past, evaluating current goods to determine their advantages and disadvantages, and speaking with specialists or prospective customers to obtain technical information. Engineers can avoid "reinventing the wheel," learn from past mistakes, and find pertinent scientific concepts or mathematical formulas like structural load requirements or material properties that will determine the success of the design by carrying out in-depth study.

1.8.3 Imagine: Develop Possible Solution

The objective is to investigate a set of strategies. After a vast number of ideas are produced, the team evaluates them in light of the previously established requirements and limitations to identify the most practical ideas. Before any actual construction starts, this step frequently produces sketches or more likely drawings that assist to conceptualize how various components could operate together.

1.8.4 Plan: Select a Promising Solution

During the Plan stage, the team chooses the most promising solution to go from a long list of ideas to a single, to a better way ahead. A Decision plan, a tool that will be used to the project's particular requirements, like cost-effectiveness, is frequently used to make this choice. Before any resources are used for actual construction, this phase acts as the project's final plan, ensuring that all team members are aware of exactly what has to be built.

1.8.5 Create: Build a Prototype

The main goal of the Create stage is to create a prototype that is using less expensive or more readily available materials in order to save time and money. This phase's main objective is to transform abstract plans into a workable, testable functional model. It is meant to be changed or even broken during the process.

1.8.6 Test and Evaluate the Prototype

During the Test and Evaluate phase, The team repeatedly tests their prototype to understand how it functions in practical settings. In order to determine where the design succeeds or fails, this phase is less about making the product work and more about collecting data and quality input such as user comfort and ease of use. The team can objectively assess if the solution is effective, and ready for production by comparing these outcomes to the initial limitations and objectives, or whether certain flaws necessitate starting over.

1.8.7 Improve: Redesign as Needed

The team uses the information collected from testing to improve their designs during the Improve stage. This is the iterative core of the process, where the team examines inefficiencies or failures to identify adjustments that would improve the final result. Since designs are rarely flawless on the first attempt, the team goes back to earlier stages such as coming up with fresh concepts or modifying the plan to address particular problems. Until the solution satisfies all initial requirements of the client and operates. It continues a cycle of development, testing, and improvement and keeps going until it reaches the quality they need.